

Advanced (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A water tank can be filled at a rate modeled by the function $f(x) = 2x$, where x is the time in hours. If the tank is filled at this rate for 2 hours, then the filling rate is reduced by 25

Q10. A company produces widgets, and the production cost can be modeled by the function $f(x) = x^2 + 10x + 20$, where x is the number of widgets produced. If the company undergoes a vertical shift of 5 units downward due to a reduction in material costs, find the equation of the new production cost function and calculate the cost of producing 10 widgets.

Chef's Tips

- Emphasize the importance of understanding function transformations and their applications in real-life scenarios.
- Encourage students to use graphing tools to visualize the transformations and check their answers.
- Remind students to show all work and explain their reasoning in the free-response questions.